

Notes: 1) Solve any one question from each group I, II, III & IV and any one question from remaining.

2) All questions carry equal marks [10 marks].

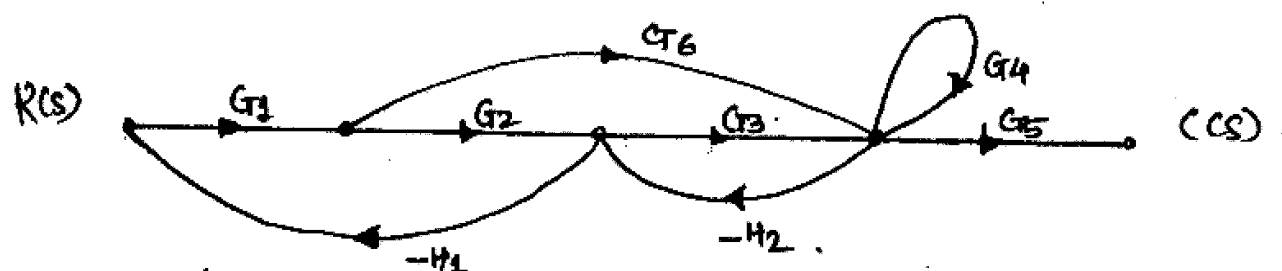
3) Draw neat labelled diagram and assume suitable data wherever necessary.

Group I

Q.1 Define the following terms: (10)

- i. Asymptotes
- ii. Centroid
- iii. Breakaway point

Q.2 Find $C(s)/R(s)$ by using Mason's gain formula. (10)



Group II

Q.3 What is root locus? Explain the rules for sketching the root locus of higher order systems. (10)

Q.4 Define open loop and closed loop control system by giving suitable examples and state the advantages of closed loop systems. (10)

Group III

Q.5 Explain the following terms: (10)

- | | |
|-------------------------------|---------------------------------|
| i) Stable system | ii) Unstable system |
| iii) Critically stable system | iv) Conditionally stable system |

Q.6 A) State Routh's criterion and also list its advantages and limitations. (05)

B) Using Routh criteria determine the stability of the system whose characteristics equation is $S^4 + 8S^3 + 18S^2 + 16S + 5 = 0$. (05)

Group IV

- Q.7 A) State Hurwitz's criterion and list its disadvantages. (05)
- B) Determine the stability of given characteristic equation by Hurwitz's method. (05)
- $F(s) = s^3 + s^2 + s + 8 = 0$
- Q.8 A) Determine the closed loop transfer function for the system shown in Figure 1. (05)

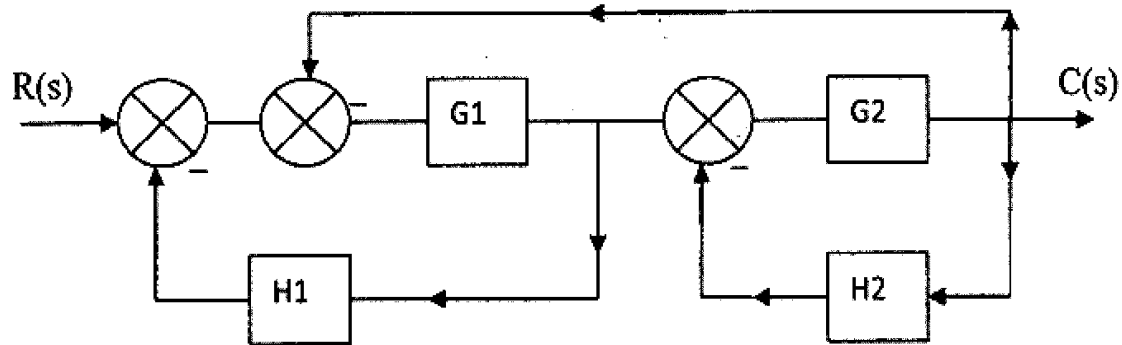


Figure 1

- B) List and explain the rules for block diagram reduction techniques. (05)

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